EXAM SYLLABUS — SQL

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1. DDL & Constraints

- CREATE TABLE syntax (columns, data types)
- PRIMARY KEY and FOREIGN KEY
- Constraints: NOT NULL, DEFAULT, CHECK, UNIQUE
- Conceptual difference between DELETE, TRUNCATE, and DROP

2. Filtering Fundamentals

- WHERE clause with comparison operators (=, <>, <, >, <=, >=)
- BETWEEN for range filtering
- IN for matching a list of values
- LIKE with wildcards (%, _) for pattern matching
- Handling missing data correctly: IS NULL / IS NOT NULL
  (and why "= NULL" does NOT work)
- ORDER BY (single and multiple columns, ASC/DESC)
- TOP N to limit result rows

3. Aggregation & Grouping

- Aggregate functions: COUNT, SUM, AVG, MIN, MAX
- GROUP BY to summarize data by category
- HAVING to filter grouped/aggregated results
- Difference between WHERE (filters rows) and HAVING (filters groups)
- Combining a JOIN with GROUP BY/HAVING

4. Joins

- INNER JOIN — only matching rows from both tables
- LEFT JOIN — all rows from the left table, matched or not
- RIGHT JOIN — all rows from the right table, matched or not
- SELF JOIN — joining a table to itself (e.g. hierarchy/reporting
  relationships)
- Table aliasing in joins

5. Window Functions

- ROW_NUMBER() — assigning a sequence number within a partition
- RANK() vs DENSE_RANK() — understanding how they differ when there are ties
- PARTITION BY and ORDER BY within OVER()
- Running totals using SUM() OVER (PARTITION BY ... ORDER BY ...)

6. Built-in Functions & CASE

- CASE WHEN ... THEN ... ELSE ... END for conditional categorization
- (Review general string/date functions like UPPER, LOWER, LEN, DATEDIFF, DATEADD, GETDATE for conceptual understanding